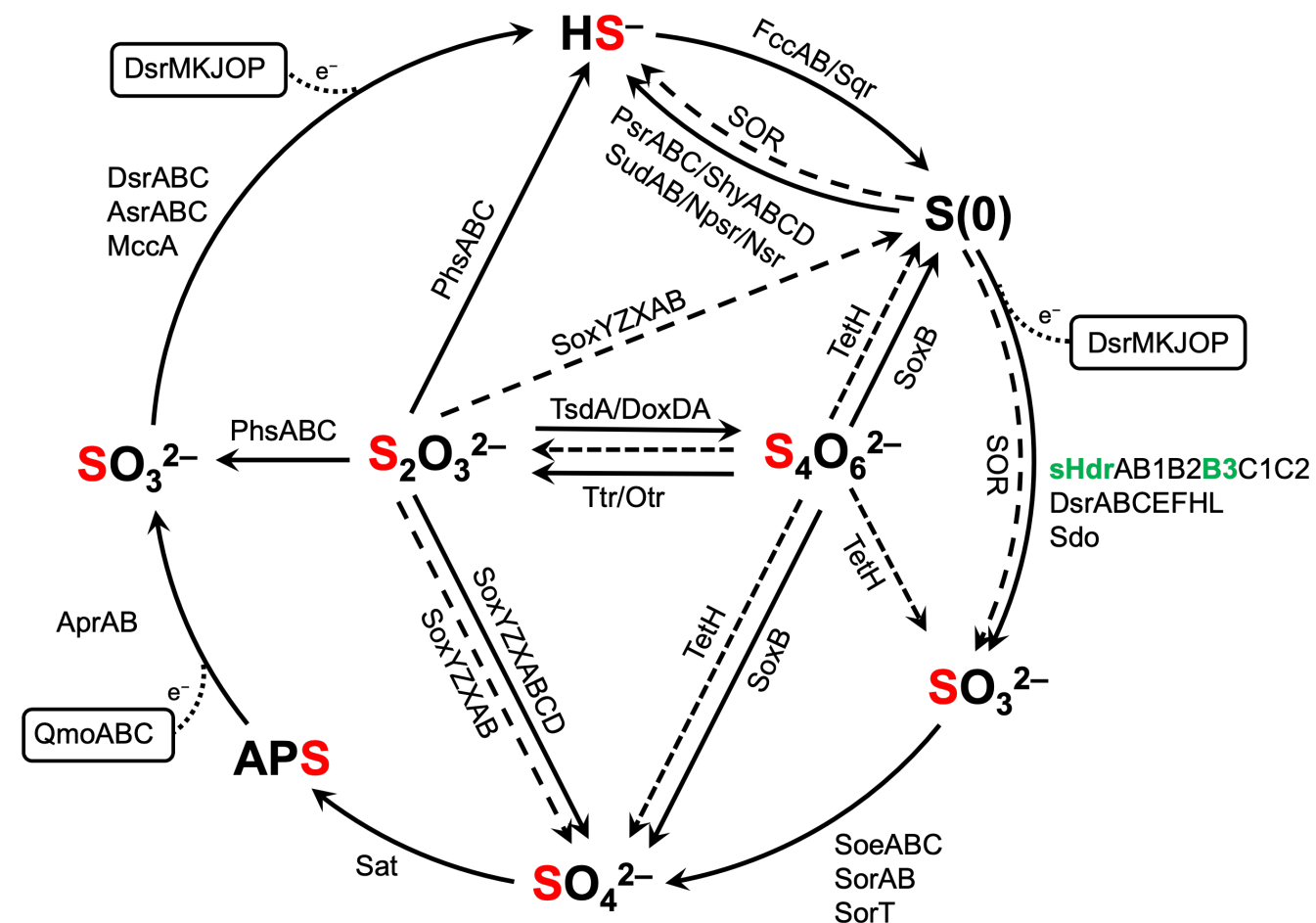


Kumpel et al. (2024)

Kumpel et al. (2024)

Cytoplasm



The diagram illustrates the metabolic pathways of sulfur in the periplasm, membrane, and cytoplasm. Key components include:

- Periplasm:**
 - Sox pathway:** SoxB and SoxCD are involved in the oxidation of H_2O to SO_4^{2-} and 2H^+ . SoxYZ is a central component that can be modified with S^- or $\text{S}_2\text{O}_3^{2-}$. SoxAX is involved in the reduction of $\text{S}_2\text{O}_3^{2-}$ to S^- .
 - Sor pathway:** SorAB and SorT are involved in the oxidation of SO_3^{2-} to SO_4^{2-} .
 - TsdA:** TsdA is involved in the oxidation of $2\text{S}_2\text{O}_3^{2-}$ to $\text{S}_4\text{O}_6^{2-}$.
- Membrane:**
 - SoeABC and DoxDA:** These are involved in the reduction of MQ to MQH_2 .
- Cytoplasm:**
 - Dsr and sHdr:** Dsr is involved in the reduction of R-S-S^- to HSO_3^- . sHdr is involved in the reduction of R-S-S^- to SO_4^{2-} .

The diagram also shows the structure of the **sHdr** gene family, divided into **type I** and **type II** genes. Type I genes include sHdrB1, sHdrB2, sHdrC1, sHdrC2, sHdrA, sHdrA, sHdrH, and sHdrI. Type II genes include sHdrB1, sHdrB3, sHdrC1, sHdrA, sHdrA, and sHdrH.